Current Biology 18, 814–818, June 3, 2008 ©2008 Elsevier Ltd All rights reserved DOI 10.1016/j.cub.2008.04.071

Report

The Cerebellum Updates Predictions about the Visual Consequences of One's Behavior

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Summary

Each action has sensory consequences that need to be distinguished from sensations arising from the environment. This is accomplished by the comparing of internal predictions about these consequences with the actual afference, thereby isolating the afferent component that is self-produced [1–4]. Because the sensory consequences of actions vary as a result of changes of the effector's efficacy, internal predictions need to be updated continuously and on a short time scale. Here, we tested the hypothesis that this updating of predictions about the sensory consequences of actions is mediated by the cerebellum, a notion that parallels the cerebellum's role in motor learning [5–8]. Patients with cerebellar lesions and their matched controls were equally able to detect experimental modifications of visual feedback about their pointing movements. When such feedback was constantly rotated, both groups instantly attributed the visual feedback to their own actions. However, in interleaved trials without actual feedback, patients did no longer account for this feedback rotation—neither perceptually nor with respect to motor performance. Both deficits can be explained by an impaired updating of internal predictions about the sensory consequences of actions caused by cerebellar pathology. Thus, the cerebellum guarantees both precise performance and veridical perceptual interpretation of actions.

Results

Motor behavior has implications for perception because our actions are a rich source of sensory information. This self-produced information needs to be distinguished from sensory events caused by external sources. It is thereby ensured that we are able to attribute self-agency to self-produced information [1, 4, 9, 10] and that this information is not perceived as part of the external environment [4, 11, 12]. In order to isolate external sensory stimuli, the afference is compared with an internal prediction about the sensory consequences of one's own behavior. In the case of a match, the afference is interpreted as a result of our own actions. In the case of a mismatch, the difference corresponds to an external sensory event [1, 13, 14]. Importantly, internal predictions build on signals related to movements, such as an efference copy of the motor command [1, 2] or proprioception. However—unlike the latter information—internal predictions must consider the fact that the same movement can have different sensory consequences: movements are performed in variable sensory environments, with sensory systems that can alter and with effectors whose efficacy can change. Hence, the perceptual consequences of self-motion can only be assessed by an internal predictor reflecting these ever-changing properties. Two reasons suggest that that the cerebellum might be the ideal substrate to form such internal predictions. First, an internal predictor requires knowledge about the efficacy of behavior. Such knowledge is available in an organ intimately involved in motor control [15, 16]. Second, internal predictions need adjustment in a way that is similar to the adjustment afforded by cerebellar motor learning [5–8, 17] and that therefore could depend on comparable computational principles.

Human imaging experiments and recordings from the cerebellum-like structure of electric fish have lent support to this notion by identifying cerebellar activity that correlated with predictions about the sensory consequences of motor acts [18, 19] and by implying that this correlate is adaptable [20, 21]. However, there remains an open question of whether the cerebellum just exhibits correlated activity or is causally involved in the representation of sensory predictions and—in the latter case—whether it generates or optimizes these predictions. Finally, if the cerebellum would optimize internal predictions, would there be a transfer of learning from the perceptual domain to the motor domain?

To address these questions, we studied patients with cerebellar lesions and their age-matched controls (Table S1, available online). In a series of two experiments, subjects were required to perform out-and-back pointing movements in a virtual-reality setup in complete darkness (Figure 1A; refer to the Supplemental Experimental Procedures, available online). This setup allowed subjects to be provided with visual feedback about the position of their index finger in real time. Feedback was accurate, altered, or lacking. The first experiment tested whether the two groups (n = 14) could predict the visual consequences of pointing. The second experiment tested whether for constantly altered visual feedback both groups (n = 12; each group) were able to update internal predictions about the new visual consequences of their behavior and, moreover, to recruit these updated predictions to optimize motor performance.

Experiment 1

Subjects were instructed to perform pointing movements in the table plane with their right index finger and to move as straight and quickly as possible (Figure 1B). Pointing distance was indicated by a briefly flashed circle (300 ms), which was centered on the starting position of the hand (9° radius). There was no visual target for the pointing movement. Rather, subjects were free to choose any position on the upper right segment of the circle as a goal for their reaches. The position of the index finger was fed back to the subjects throughout each trial. Feedback was rotated by varying degrees around the starting point of the movement in a clockwise or counterclockwise direction. After having completed their pointing movement, subjects reported the direction of the perceived rotation of visual

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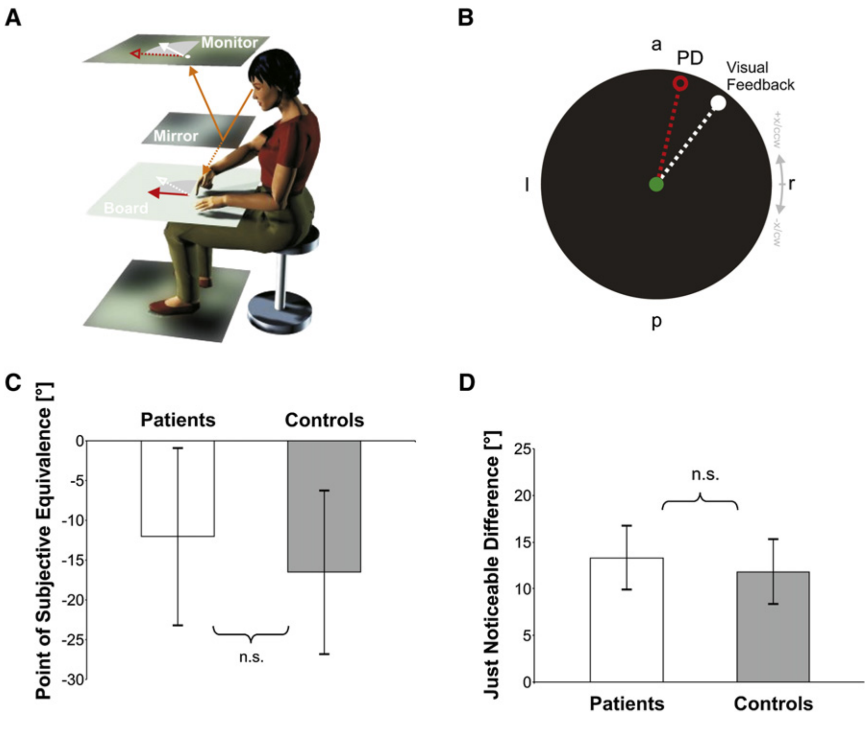

Figure 1. Accuracy of Internal Predictions

(A) Experimental setup. Subjects viewed the virtual image of their finger on the feedback monitor via a mirror (solid orange line) while performing out-and-back pointing movements with their right hand. The virtual image appeared as lying in the same plane as their actual finger (red solid arrow: actual movement vector; white solid arrow: rotated visual feedback).

(B) Paradigm of experiment 1. Online visual feedback (white disc) about pointing (red circle) was either rotated in a counterclockwise (ccw) or—as in this example—in a clockwise (cw) direction. Dotted lines indicate the movement trajectory and the respective feedback. PD denotes the actual pointing direction, and a, p, l, and r denote anterior (90°), posterior, left, and right (0°), respectively. Subjects' task was to indicate the direction of rotated visual feedback in a two-alternative forced-choice manner by pressing one of two buttons with their left hand.

(C and D) Group results. There was no significant difference between groups with respect to the point of subjective equivalence between the actual and the perceived pointing directions (C) and the just noticeable difference between these perceived directions (D). Both results indicate a preserved ability of patients to detect visual feedback manipulations by resorting to intact internal predictions for familiar actions. Bars depict means calculated across subjects (± 95% confidence intervals, uncorrected).

feedback with respect to their actual movement in a two-alternative forced choice manner: they reported their decision by pressing one of two buttons with either the left index finger (clockwise) or the left middle finger (counterclockwise). The amount of rotation in individual trials was determined by three randomly interleaved staircase procedures [22]. Two procedures estimated the detection thresholds (75% hits) for perceived clockwise and counterclockwise feedback manipulations, respectively. The "just noticeable difference" of feedback deviation reflects the mean of these two detection thresholds. A third staircase procedure was used to titrate the "point of subjective equivalence" between perceived movement and visual feedback. At this point subjects responded at chance level (Figure S1). Because subjects' choices critically depended on a comparison of the actual visual feedback with the predicted visual action outcome, both the point of subjective equivalence and the just noticeable difference can be considered indirect measures of the accuracy of internal predictions.

Both groups showed a clockwise bias of the point of subjective equivalence toward the side of the effector. More importantly, between both groups there was no significant difference in the point of subjective equivalence (see Figure 1C) or in the just noticeable difference (see Figure 1D). Finally, the variability of the two perceptual estimates was identical in the two groups (F test). In summary, experiment 1 indicates that patients were impaired neither in their generating of accurate internal predictions nor in their matching of them with visual feedback (e.g., secondary to potential oculomotor deficits [23]).

Experiment 2

In experiment 2, we asked whether subjects were able to perceptually adapt to altered sensory consequences of their pointing movements by recalibrating internal predictions. Moreover, we tested whether they would exploit updated internal predictions to optimize motor performance. As in experiment 1, the basic task of subjects was to carry out pointing movements with their right index finger. After each movement, they had to give a perceptual estimate about their perceived pointing direction by placing a mouse-guided cursor into the respective direction with the use of their left hand. Three different experimental conditions were presented in a randomly interleaved manner. (1) During "feedback trials," subjects received visual feedback about their movements, which induced and maintained adaptation. This condition allowed us to compare subjects' perception of self-action (indicated by their perceived pointing direction) with their actual motor behavior (indicated by their actual pointing direction) while both external visual feedback and internal predictions about the expected feedback were available (see Figure 2A). During the preadaptation phase, this feedback was kept veridical to get a baseline estimate of the perceived pointing direction. In the "adaptation buildup phase," we gradually increased the feedback rotation angle in steps of -6° (clockwise) over five consecutive trials. After this build-up, rotation was kept constant at -30° (clockwise). This postadaptation phase allowed us to maintain adaptation. Two types of probe trials were randomly interleaved with the feedback trials, the first type tested for recalibration of internal predictions ("perceptual probe trials") and the second type for changes in motor performnce ("motor probe trials"). For further illustration of the course of experiment 2, please refer to Figure 2D. (2) The perceptual probe trials were identical to feedback trials except that no visual feedback was provided at all (see Figure 2B). This condition allowed us to detect whether the new visual feedback in feedback trials was used to update internal predictions about pointing direction, since, in the absence of visual feedback, visual estimates of perceived pointing critically depended on internal predictions. (3) During motor probe trials, subjects

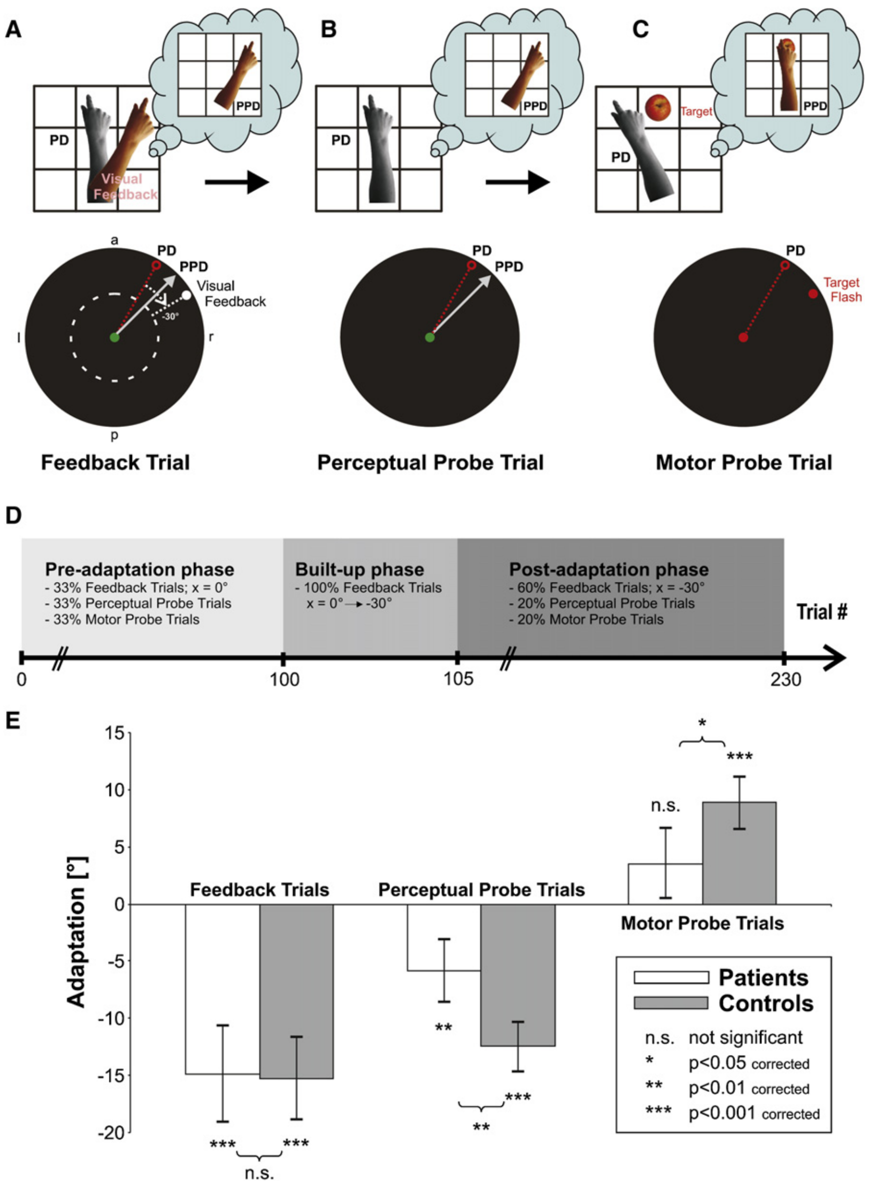

Figure 2. Updating of Internal Predictions

(A–D) Experimental conditions of experiment 2. The top row of panels A–C shows a graphical illustration of each of our experimental conditions. These illustrations also indicate the expected effects of adaptation on a control population [9]: if subjects were constantly provided rotated visual feedback ([A], colored arm) with respect to their actual pointing movements (gray arm), controls would tend to perceive their arm as pointing into the direction of visual feedback. Even without visual feedback (B), controls would continue to perceive their arm as pointing into the same visual direction (A). They would have internalized the new visual consequences associated with this particular action. Finally, if asked to reach for a specific visual target (apple) without visual feedback (C), controls would reach in a direction opposite to the feedback rotation in order to account for the altered internal prediction about the visual consequences of their movement. If the cerebellum were to play a key role in the updating of such internal predictions, one would expect to see a specific deficit of patients in the updating of both their perceived pointing direction (PPD) and their actual pointing direction (PD) in conditions that critically depend on accurate internal predictions, i.e., in (B) and (C), respectively. The bottom row of panels A–C depicts the three different experimental paradigms that were used to drive adaptation and to assess its effects on both perception (PPD) and action (PD). Conditions were presented in randomly interleaved trials, in which subjects performed out-and-back pointing movements (exemplary pointing direction, dotted red line). After each movement subjects had to indicate their perceived pointing direction with the use of a mouse-guided visual cursor (solid gray arrow). (A) In feedback trials subjects received veridical online visual feedback about the peripheral part of their hand movement within a first block of 100 trials, i.e., the preadaptation phase. Afterward, feedback was spatially rotated in order to induce adaptation. The feedback angle *x* was gradually increased during the buildup phase and remained constant during the postadaptation phase ($x = -30^\circ$; white dotted line). Visual feedback of the first half of the trajectory was blocked by an invisible black mask of 4.5° diameter (white dashed circle) to diminish subjects' overall exposure to feedback and to limit the possibility of online motor corrections (for detailed discussion, see [9]). (B) In perceptual probe trials, subjects received no visual feedback at all. (C) In motor probe trials, an explicit target (red circle) was briefly flashed and subjects had to perform a pointing movement toward the remembered target location, again while no visual feedback was present. A detailed time course of experiment 2 is depicted in (D), which indicates the share of each experimental condition and also specifies *x*, namely the angle of visual feedback rotation, for each of the different stages of adaptation.

(E) Group results. Both patients and controls showed a comparable, highly significant adaptation of their perceived pointing direction as long as visual feedback was present (feedback trials). However, in the absence of visual feedback, this adaptation of the perceived pointing direction was significantly smaller in patients than in controls (perceptual probe trials). Moreover, patients showed a significantly reduced compensatory motor adjustment (motor probe trials; figure conventions as in Figure 1).

made a pointing movement toward a visual target that was briefly flashed randomly at one of four possible positions (90°, 60°, 30°, or 0°; see Figure 2C). Because no visual feedback was given, this condition allowed us to test whether subjects could internally guide pointing in such a way that it would account for the new visual consequences. Figures 2A–2C provide graphical illustrations of the parameters tested and depict the expected effects of adaptation on each of these parameters [9].

When visual feedback was constantly altered in the postadaptation phase, both groups showed a significant shift of their perceived pointing direction in feedback trials as compared to the preadaptation phase (see Figure 2E; also, refer to Figure S2 for individual examples). Because there was no significant difference in the size of the perceptual shift between groups, both were equally able to draw on visual feedback to adjust perception of self-action.

However, when visual information was absent (perceptual probe trials) both groups still adapted, but the change of perceived pointing direction was significantly smaller in patients as compared to controls (see Figure 2E). In the control group, the perceptual changes in feedback trials and in perceptual probe trials were statistically indistinguishable, thus demonstrating that the altered perceptual estimate builds on an internal prediction about the expected visual consequences of pointing, a prediction that got recalibrated by new visual feedback [9]. In contrast, patients adapted significantly less in perceptual probe trials as compared to feedback trials—a perceptual deficit that reveals the insufficient recalibration of their internal predictions. Importantly, given that patients' estimates of the perceived pointing direction during the postadaptation phase were stable and did not show any residual buildup (regression analysis; $p > 0.05$), this perceptual deficit cannot simply be explained by an assumption of longer time constants for learning in the patient group.

Although feedback trials required subjects to develop a coherent percept of self-action despite conflicting internal and external information about movement, they did not require subjects to adapt motor performance. However, controls exhibited a compensatory adjustment of pointing in motor probe trials despite the lack of visual feedback, indicating that they "optimized" motor performance by recruiting updated internal predictions. Specifically, the difference between the actual pointing direction and the position of the flashed target showed a significant, adaptation-induced increase, comparable in size but opposite in direction to the changes in the perceptual probe trials (see Figure 2E). The patient group, however, did not show a significant motor adjustment, and the amount of change was significantly smaller. These findings reveal the failure of patients to modify motor control by the use of updated internal predictions about the visual consequences of action.

Finally, there is preliminary evidence that the learning deficits observed in this study are caused by disorders of cerebellar domains ipsilateral to the effector. In two patients with unilateral cerebellar lesions who were subsequently tested with both their right and their left hand, the results for the ipsilesional side and the contralesional side qualitatively resembled the results for patients and controls, respectively (see Figure S3).

Discussion

Our results extend the classical view of the cerebellum as a site of motor learning by showing that it likewise is a major site of perceptual learning. More specifically, they suggest that the cerebellum is responsible for rapidly updating internal predictions about the visual consequences of motor behavior in order to inform the perception of self-action.

Patients' ability to predict the visual consequences of a well-learned task (experiment 1) ruled out the possibility that a more general sensory or motor deficit could account for our results. Moreover, the fact that patients were able to draw on visual feedback to estimate their own behavior (experiment 2) shows an intact integration of conflicting visual and proprioceptive cues for the multimodal interpretation of self-motion. Only when visual feedback was no longer available did patients differ from controls, in that they were impaired in (1) the updating of a visual estimate of the direction of pointing and in (2) adjustment of their movements accordingly.

Specifically, in comparison of the preadaptation and the postadaptation phases, one and the same movement was perceived differently by the controls. This was indicated by a shift of the perceived pointing direction that occurred whether or not visual feedback was present. Given that both proprioceptive inputs and motor commands were the same for the same movement, it could not be a change in proprioception or in efference copy per se that accounted for this altered interpretation of self-motion. Hence, there must be a change in an internal representation that relates these internal sources of self-motion information to the new visual consequences of pointing, namely a plastic internal predictor [9]. The significantly reduced modification of patients' perceived pointing direction in trials without visual feedback thus reflects a specific deficit in updating of internal predictions, which is caused by cerebellar pathology.

With respect to motor control, patients and monkeys with cerebellar lesions still exhibit adequate motor behavior in tasks that they have been well trained to perform [7], suggesting some compensatory mechanisms in the case of chronic lesion [5, 24]. In contrast, both species show deficits in the adapting of behavior on a short time scale [5, 7, 8]. As shown here, the same principles apply for internal predictions supporting the perception of self-action. Thus, our findings might not only specify cerebellar contributions to perceptual and cognitive functions [25–27] but also support the idea of the cerebellum's uniform contribution to both perception and action: the cerebellum integrates motor and sensory information in order to predict the consequences of behavior—not only to fine tune movement (e.g., [5–7]) but also to inform the perception of self-action [3, 16, 28]. This is achieved by exploiting of the difference between a predicted-state estimate and an actual-state estimate, which results in a prediction error. The cerebellum acts to reduce this error by recalibrating the internal prediction on a short time scale. In other words, the cerebellum forms a "new" or "updated" internal model of the consequences of behavior [3, 6, 16], whereas the long term representation of internal models might be realized elsewhere [21, 29].

Internal predictions could serve as "internal feedback signals" for action [16] and for action perception. They would allow precise perceptual estimates and performance even in situations with delayed or missing sensory feedback. If internal predictions were not optimized accordingly, both perception and action would no longer be accurate in such situations. In fact, patients were able to provide accurate perceptual estimates, but only as long as visual feedback was available and as long as they didn't have to rely on updated internal predictions (experiment 2).

We further assessed changes in subjects' motor performance in experiment 2. In randomly interleaved motor probe trials subjects had to point at a flashed visual target. However, there was no visual feedback that would have allowed subjects to estimate pointing accuracy in order to optimize their behavior. Thus, any adaptation-induced change of pointing could not be directly explained by a "visual error signal" or any corrective movement [30], which were both lacking. Rather, subjects could adjust performance solely on the basis of updated internal predictions. Control subjects showed an adaptation-induced change in pointing direction that was equal in size but opposite in direction to the changes in perceived pointing direction. Hence, they performed a compensatory movement that exactly accounted for the new visual action consequences captured by their internal predictions. Accordingly, patients were impaired in the optimization of motor control because of the lack of updated internal sensory predictions. This is further corroborated by the fact that an altered perceptual interpretation of self-action in the presence of visual feedback was not sufficient to alter motor behavior per se. Our hypothesis is in good agreement with recent studies suggesting that a sensory predictor could be used to train a motor controller [31, 32] and that a cerebellar prediction of the visual consequences of movement might be used to guide motor planning

[19]. The fact that both action and self-action perception seem to build on these updated predictions implies a major ecological importance and a rather universal character of the cerebellar mechanism described.

Supplemental Data

Experimental procedures, patients' characteristics (Table S1), and figures illustrating the course of experiment 1 (Figure S1) and the time course of adaptation in two exemplary subjects (Figure S2), as well as the effects of lateralized cerebellar lesions in experiment 2 (Figure S3) are available with this article online at http://www.current-biology.com/cgi/content/full/18/11/814/DC1/.

Acknowledgments

We thank Josh Wallman for his helpful comments on earlier versions of this paper. This work was supported by grants from the Volkswagen-Stiftung (VW I/80 727 to P.T. and European Platform to M.S.) and from the German Research Council Collaborative Research Centre 550 (P.T.).

Received: February 20, 2008
Revised: April 23, 2008
Accepted: April 23, 2008
Published online: May 29, 2008

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Supplemental Data

The Cerebellum Updates Predictions about the Visual Consequences of One's Behavior

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Supplemental Experimental Procedures

Subjects

The patient group consisted of fourteen patients (ten men, four women; average age 52 yrs). They suffered either from global cerebellar degeneration with no clinical, MRT, or electrophysiological evidence for extracerebellar manifestations of the disease (two individuals with SCA 6; six individuals with isolated cerebellar degeneration of unknown origin; all genetically negative for SCA 1, 2, 3, 6, 7 or 17) or from circumscribed cerebellar lesions due to stroke or tumors (six stroke patients, studied 2–5 days after stroke manifestation). None of the patients was suffering from a variant of multiple-system atrophy or from SCA other than SCA 6. All subjects were right-handed. In the case of unilateral damage, we considered only patients with lesions ipsilateral to their preferred hand for the group comparisons. Two patients (HELO and EDBR) did not participate in the second experiment. Table S1 summarizes the relevant information about all patients (gender, age, etiology of the disease, location of the lesion, ataxia severity), and it also indicates the experiments in which each patient had participated.

Fourteen control subjects, matched for age and handedness, participated in the first experiment (six men, eight women; average age 52 yrs). Two control subjects, matched with the two patients that did not participate in the second experiment, were studied in experiment 1 only. The majority of control subjects (n = 12) were inpatients suffering from lumbar radiculopathies without any affection of supralumbar structures and without medication affecting the CNS. By using lumbar radiculopathy patients as controls, we tried to account for unspecific effects (e.g., hospitalization, general disease experience) that might have affected the required perceptual evaluations. The remaining two control subjects were two healthy individuals. All subjects had normal or corrected-to-normal visual acuity and gave their informed consent according to the declaration of Helsinki.

Experimental Setup

Subjects were seated in front of a large horizontal board with their heads stabilized in a head-and-chin rest. They looked down onto a rectangular mirror placed horizontally halfway between the board and a computer screen, which was fixed above the mirror. Subjects always viewed the stimulus screen via the mirror. For geometrical reasons this screen appeared as lying in the plane of the tabletop (see Figure 1A). Both hands of the subjects were placed on the board, below the mirror, and thus were invisible to them. Furthermore, we prevented orientation clues from the environment by carrying out the experiments in complete darkness. On the top of the subjects' right index finger an ultrasound emitter was mounted. Its position was recorded by a 3D real-time motion-tracking system (Zebris CMS 70 P; Isny, Tübingen; Germany). Positional information was transferred to the stimulus computer in order to feed back the position of the index finger visually via the monitor-mirror device. The feedback stimulus consisted of a 0.4° diameter gray disc that appeared to be in spatial correspondence with subjects' index-finger tip as long as we kept the feedback veridical. The position of the disc was updated online at a frame rate of 60 Hz.

Table S1. Etiology of Disease, Lesion Description, and Ataxia Severity

Demographics			Lesion Description		Ataxia Rating						Experimental Involvement		
Patient Identification	Sex	Age	Etiology	Location	PO (34)	LL (24)	LR (24)**	SP (8)	OM (6)	Total (96)**	Hand Tested	Exp. 1	Exp. 2
1 UTBÜ	F	38	EOCA	Global	3	6	7	1	6	23	R	+	+
2 WOFI	M	60	AT	Global	9	5	5	3	4	26	R	+	+
3 GEFO	M	56	ADCA	Global	8	7	6	2	2	25	R	+	+
4 HELO	M	68	ADCA	Global	10	10	8	2	6	36	R	+	–
5 GEHO	M	63	SCA 6	Global	12	7	7	4	6	36	R	+	+
6 KAFU	F	74	SCA6	global	14	6	7	2	5	34	R	+	+
7 BEWI	M	43	AT	almost exclusively vermis	8	6	6	1	1	22	R	+	+
8 EDBR	F	59	AT	Global	15	11	9	4	2	41	R	+	–
9 BELU	M	38	Right superior cerebellar artery infarction	Right cerebellar hemisphere + anterior vermis	5	2	2	4	2	15	R	+	+
10 NOWE	M	48	PICA right	Inferior right hemisphere	3	2	2	3	2	12	R	+	+
11 EBHE	M	40	PICA right	Right hemisphere	5	3	3	1	2	14	R	+	+
12 JELO	M	66	PICA both sides	Lesion right > > left hemisphere	10	4	5	1	2	22	R	+	+
13 TOWO	M	37	Ependy-moma right	Right hemisphere + median cerebellum	12	4	7	1	1	25	R	+	+
14 ANNO	F	44	PICA right	Right hemisphere	4	2	5	0	1	12	L/R	+	+
15 SILO	F	33	Ganglio-cytoma left	Left hemisphere + median cerebellum	9	7	3	1	1	21	L/R	–	+*

EOCA denotes early onset cerebellar atrophy; AT denotes cerebellar atrophy of unknown origin; ADCA denotes autosominal-dominant cerebellar atrophy; SCA denotes spinocerebellar atrophy; PICA denotes infarction of the posterior inferior cerebellar artery.
M denotes male; F denotes female.
R denotes right hand; L denotes left hand.
PO, posture; LL and LR, limb ataxia left and right (including lower and upper limb), respectively; SP, speech disorders; OM, oculomotor disorders.
The score that would indicate maximal impairment on each subtest is given in parentheses.
* Patient SILO was not included in the group analysis of experiment 2 because her lesion was contralateral to her preferred right hand.
Ataxia rating was performed according to the International Cooperative Ataxia Rating Scale [S4].
** For the sake of a symmetric assessment of limb ataxia, the subtest 14 (drawing) was dropped from the evaluation.

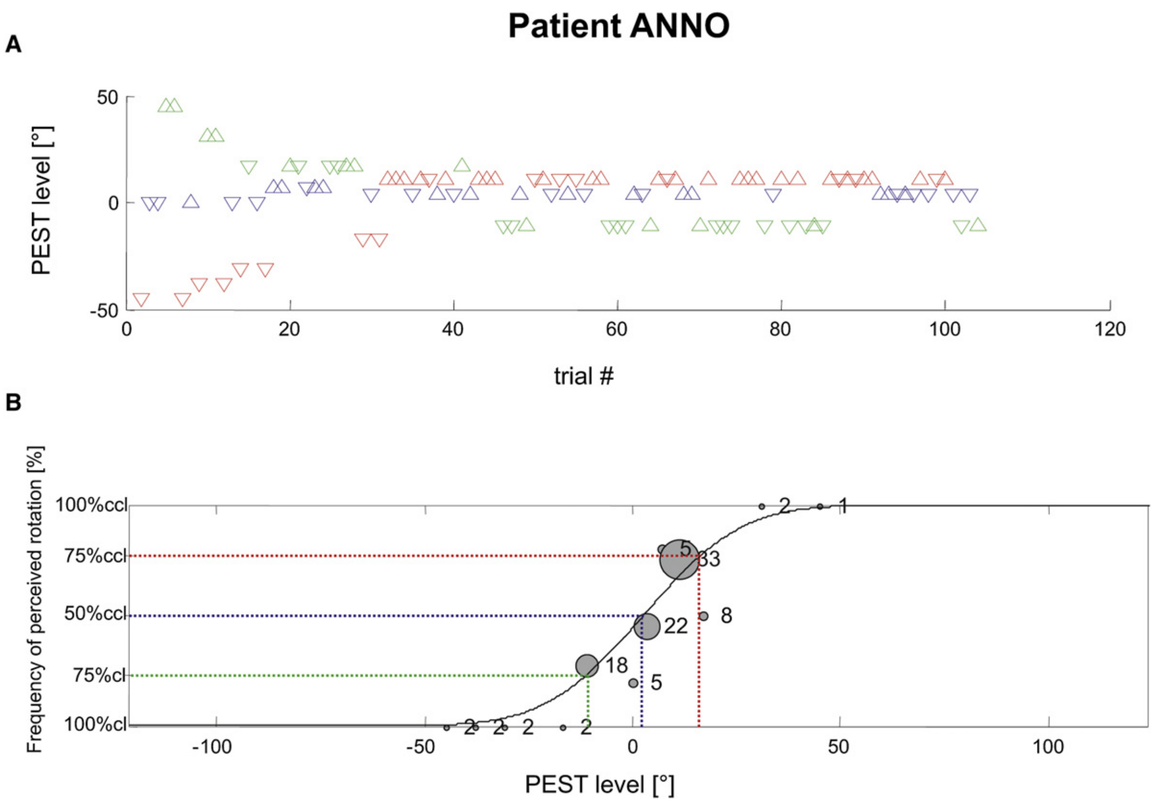


Figure S1. Experiment 1: Exemplary Results of a Patient

(A) Course of experiment 1. Subjects' task was to detect the direction of rotation of visual feedback (clockwise or counterclockwise) with respect to their actual hand movement: Each triangle represents a single trial during experiment 1, and the PEST (Parameter Estimation by Sequential Testing [S2]) level of each triangle indicates the amount and direction of the visual feedback rotation imposed in particular trials. The orientation of the triangles represents the individual perceptual decision made by the subject with the use of the button box (upward triangles indicate perceived counterclockwise rotation; downward triangles indicate perceived clockwise rotation). The amount of visual feedback rotation was determined by three randomly interleaved staircase procedures according to the PEST algorithm (red, blue, and green triangles). Two staircase procedures were employed to estimate the detection thresholds (75% hits) for perceived clockwise (green triangles) and perceived counterclockwise (red triangles) feedback manipulations. The third staircase procedure was used to estimate the point of subjective equivalence between hand movement and visual feedback (point of subjective equivalence, 50% clockwise decisions; i.e., chance level; blue triangles).

(B) Probit analysis. Precise perceptual threshold estimates were obtained for each subject by the running of a Probit analysis [S3]. This analysis was calculated across all individual data points that were obtained by the staircase procedures around the thresholds of interest. The number of samples is depicted next to each data point and is also reflected by its size. The Probit function fitted to the data represents the frequency of perceived counterclockwise-rotations for different PEST levels. The absence of perceived feedback rotation is indicated by 50% clockwise and 50% counterclockwise decisions. The point of subjective equivalence in this example was 2.8° (see blue dotted lines). Detection thresholds were calculated as the average amount of clockwise and counterclockwise- rotation needed in order to be perceived with a likelihood of 75% (green and red dotted lines, respectively). In this particular example we yielded an average detection threshold of 12.5°. We refer to this estimate as the "just noticeable difference" of feedback rotations.

In all experiments subjects had to perform out-and-back pointing movements with their right index finger. Rapid movements were encouraged by a short trial-duration limit of 1500 ms. Subjects were instructed to perform the movements as quickly and as straight as possible. In the beginning of each trial subjects had to place their right index finger on the center of the board, which was defined by a tactile cue, namely a small nail head. This center location always corresponded to the position of the fixation target.

Data Analysis: Experiment 1

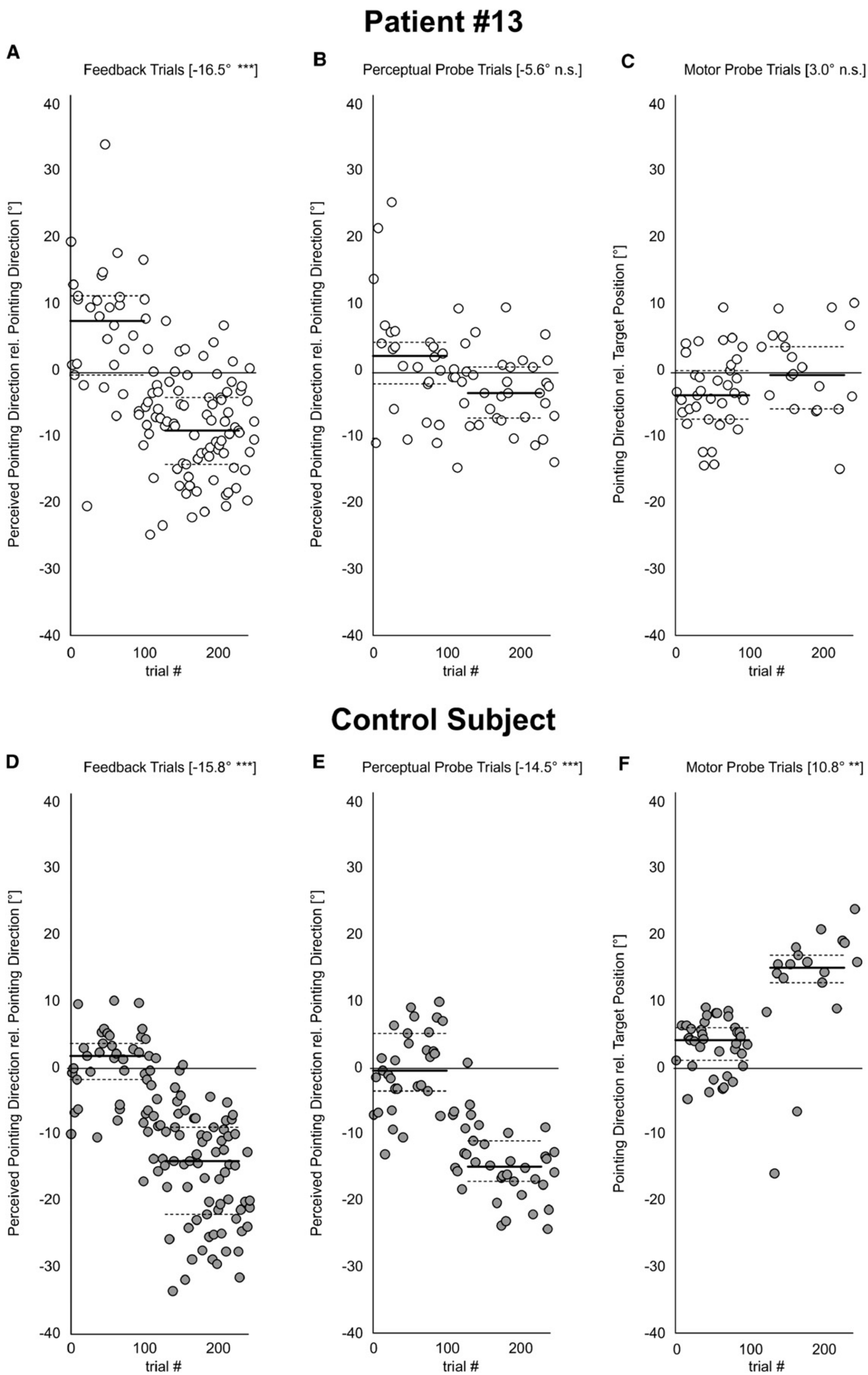
"Point of subjective equivalence" and "just noticeable difference" were compared at the group level. We tested for differences in these measures between patients and controls by employing multiple t tests (H_0: No difference between groups; two-tailed tests). Resulting p values were always corrected for multiple comparisons according to the Bonferroni procedure.

Data Analysis:Experiment 2

The trajectory of each movement was recorded and stored on computer disc for offline analysis. The manual pointing direction was defined as the slope of a linear regression calculated across the 2D x- and y- finger position samples that were acquired during the outward journey of the hand (see [S1] for details). To estimate the precision of pointing, we also calculated the difference between the angle of the actual pointing direction and the angle at which the target flash in motor probe trials had occurred.

A subjective visual estimate of subjects' center-out movement direction – the perceived pointing direction—was acquired after each trial. This was accomplished by subjects moving a mouse-guided cursor relative to the fixation spot with the left hand into a position that corresponded to their perceived pointing direction. This procedure can be compared with the placing of a watch hand as an indicator of the perceived pointing direction: Linear movements of the mouse were transferred to a circular-movement path of the cursor pointer around the fixation spot on a white circle (3° diameter). The final position of the pointer—indicating the perceived pointing direction—was confirmed by subjects pressing the left mouse button.

Effects of adaptation on the relative pointing direction (in motor probe trials) and on the relative perceived pointing direction (in feedback trials and perceptual probe trials) were studied by comparison of the differences between the pre-adaptation (trial 1–trial 100) and the post-adaptation phase (trial 106–trial 230). The first 25 trials of the post-adaptation phase were discarded in order to guarantee that adaptation had already been accomplished (compare Figure 3). In fact, we performed a linear-regression analysis over the respective behavioral measures in the post-adaptation phase in order to probe for any residual adaptation throughout this phase.


Patient #13
A
Feedback Trials [-16.5° ***]
B
Perceptual Probe Trials [-5.6° n.s.]
C
Motor Probe Trials [3.0° n.s.]
Perceived Pointing Direction rel. Pointing Direction [°]
Pointing Direction rel. Target Position [°]
40
30
20
10
0
-10
-20
-30
-40
0
100
200
trial #
Control Subject
D
Feedback Trials [-15.8° ***]
E
Perceptual Probe Trials [-14.5° ***]
F
Motor Probe Trials [10.8° **]

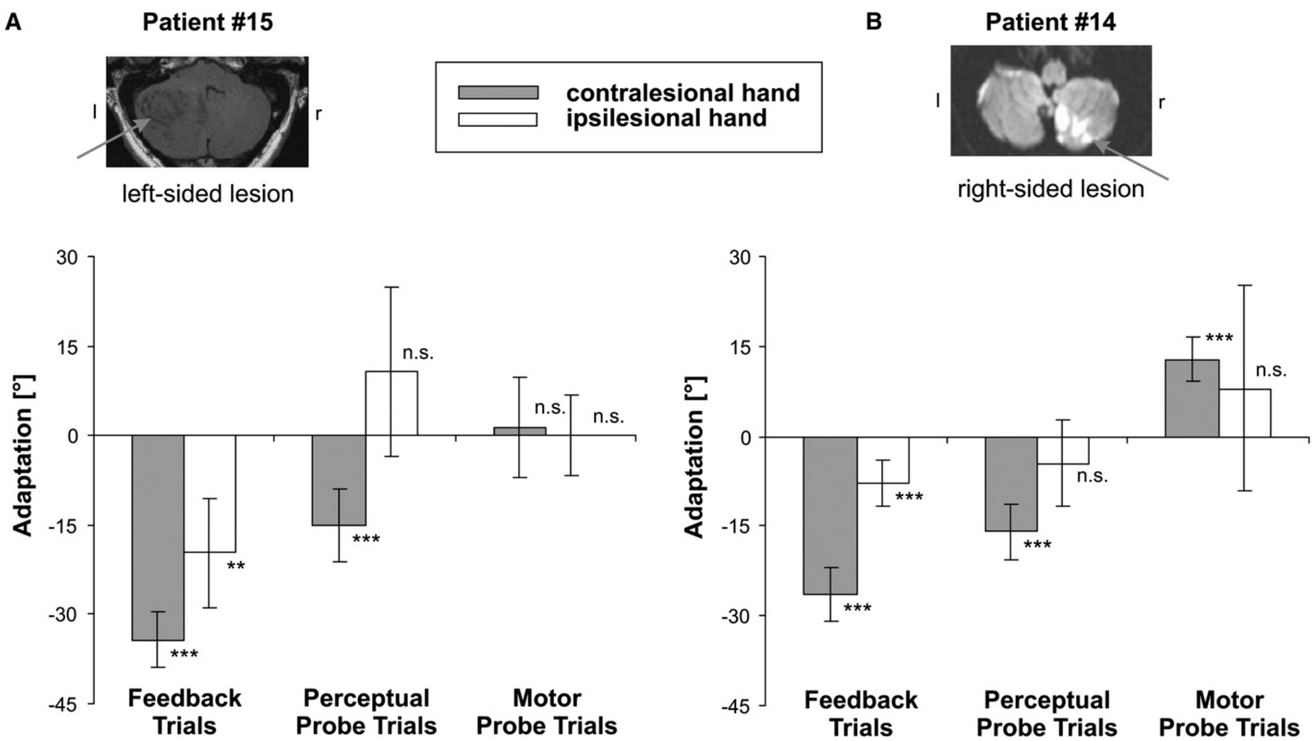


Figure S3. Perceptual and Motor Learning in the Ipsilateral Cerebellum

To test for lateralization of the observed effects within subjects, one patient ([B]; no. 14, ANNO, right-handed, female, 44 years) with a lesion of the right cerebellum additionally performed experiment 2 with her contralesional left hand. Furthermore, another patient ([A]; no.15, SILO; right-handed, female, 33 years) with a unilateral cerebellar lesion on the opposite (left) side was specifically recruited for this within-subject comparison of lateralization. Each bar depicts the mean difference (± 95% confidence interval, uncorrected) between the respective pre-adaptive and post-adaptive behavioral measures—perceived pointing direction relative to pointing direction for both feedback probe trials and perceptual probe trials, as well as pointing direction relative to target position for motor probe trials. When performing hand movements with their contralesional hand (gray bars), both patients changed their perceived pointing direction significantly in feedback trials and perceptual probe trials. One patient also significantly altered his motor behavior in motor probe trials. When using their ipsilesional hand (white bars), both patients adapted only in feedback trials; neither in perceptual probe trials nor in motor probe trials. Thus, the two patients resembled the performance of control subjects when tested with their contralesional hand and the performance of the patient cohort when tested with their ipsilesional hand.

Given the rather limited number of trials available for such an analysis within a single subject, this analysis was separately performed for each of the psychophysical measures that were pooled across the group of patients and across the group of age-matched controls, respectively.

As in experiment 1, behavioral measures were compared on the group level. In order to statistically test for adaptation within and across subjects, we performed additional t tests (H_0: no difference for pre-adaptation and post-adaptation phases; two-tailed tests). All resulting p values were Bonferroni corrected for multiple comparisons within subjects.

Supplemental References

S1. Synofzik, M., Thier, P., and Lindner, A. (2006). Internalizing Agency of Self-Action: Perception of One's Own Hand Movements Depends on an Adaptable Prediction About the Sensory Action Outcome. J. Neurophysiol. *96*, 1592–1601.

S2. Lieberman, H.R., and Pentland, A.P. (1982). Microcomputer-based estimation of psychophysical thresholds: The best PEST. Behav Res Meth & Instr *14*, 21–25.

S3. McKee, S.P., Klein, S.A., and Teller, D.Y. (1985). Statistical properties of forced-choice psychometric functions: implications of probit analysis. Percept. Psychophys. *37*, 286–298.

S4. Trouillas, P., Takayanagi, T., Hallett, M., Currier, R.D., Subramony, S.H., Wessel, K., Bryer, A., Diener, H.C., Massaquoi, S., Gomez, C.M., et al. (1997). International Cooperative Ataxia Rating Scale for pharmacological assessment of the cerebellar syndrome. The Ataxia Neuropharmacology Committee of the World Federation of Neurology. J. Neurol. Sci. *145*, 205–211.

Figure S2. Time Course of Adaptation in Two Exemplary Subjects

Both the patient and the control subject exhibited a significant change in the relation between their perceived pointing direction and their actual pointing direction before (trials 1–100) and after (trials 106–230) visual feedback was rotated in feedback trials (A and D). This change is clearly evident in a comparison of the median values (bold lines) and the nonoverlapping quartiles (thin broken lines) of the perceptual estimates between the pre-adaptation and post-adaptation phases for the patient (A) and for the control (D). In the absence of visual feedback, only the control subject showed a comparable, adaptation-induced shift of his perceived pointing direction in perceptual probe trials (E) and a compensatory counter-rotation of his pointing direction in motor probe trials (F). Despite a trend into the same direction, both measures did not significantly change for the patient (see [B] and [C], respectively). Subheadings provide further information about the size of the adaptation-induced effects, as well as their level of significance (conventions as in Figure 1).